

Product Requirements Document (PRD)

Project Name

ChatVault

WhatsApp Export Chat Viewer & Media Player

1. Product Overview

ChatVault is a web-based WhatsApp export chat parser and viewer that allows users to upload exported WhatsApp chat files and view them in a WhatsApp-style chat interface.

The app is completely read-only and offline-focused. It is NOT a messaging platform.

Users can:

- Upload WhatsApp exported ZIP or TXT files
- Parse conversations
- View chats in WhatsApp-style UI
- Play voice notes/audio inline
- View images/videos/documents
- Search chats
- Navigate large conversations smoothly

The goal is to recreate the feeling of reading a real WhatsApp chat while maintaining privacy and local processing.

2. Product Goals

Primary Goals

- Parse WhatsApp exported chats accurately
- Display chats in a familiar WhatsApp-like UI
- Support inline media playback
- Handle huge chat exports efficiently
- Work fully in-browser without server upload

3. Non-Goals

The app will NOT:

- Send messages
- Connect to WhatsApp servers
- Support real-time messaging
- Require phone number login
- Store user chats permanently on cloud
- Clone WhatsApp exactly

4. Target Users

Primary Users

- Users backing up old chats
- Users reading archived conversations
- Long-distance couples/friends preserving memories
- Researchers/journalists analyzing chats

- Users migrating between devices

5. Core Features

5.1 Chat Import

Supported Inputs

ZIP Export containing:

- TXT chat file
- audio files
- images
- videos
- documents

TXT-only export support.

5.2 WhatsApp Chat Parsing

Supported Message Types

Type	Support
Text	Yes
Emoji	Yes
Voice Notes	Yes
Images	Yes
Videos	Yes
Documents	Yes
Stickers	Optional
Reactions	Future

5.3 Chat Rendering

UI Requirements

Interface should resemble:

- WhatsApp Web
- grouped message bubbles
- timestamps
- sender grouping
- date separators

But:

- different colors
- different typography
- original branding

5.4 Audio Playback

Features

- Play/Pause
- Seek
- Progress indicator
- Playback speed

- Continue playback while scrolling
- Single active playback

5.5 Search

Users can:

- search messages
- jump between results
- highlight matched text

5.6 Large Chat Optimization

Must support:

- 100k+ messages
- virtual scrolling
- lazy rendering

5.7 Privacy Mode

All parsing occurs locally in browser.

No chats uploaded to server.

6. User Flow

Landing Page

↓

Upload ZIP/TXT

↓

Parsing Screen

↓

Media Linking

↓

Chat Rendering

↓

Search / Playback / Navigation

7. UI/UX Requirements

7.1 Layout

Desktop:

Sidebar | Chat Viewer

Mobile:

Full-screen chat viewer

7.2 Chat Bubble Behavior

Incoming Messages:

- left aligned

Outgoing Messages:

- right aligned

Grouped Messages:

- merged spacing similar to WhatsApp

7.3 Media Rendering

Images

- rounded previews
- click to expand

Audio

- inline player
- waveform
- progress

Documents

- downloadable card

8. Functional Requirements

ID	Requirement
FR-1	User can upload ZIP
FR-2	System extracts ZIP locally
FR-3	System parses TXT file
FR-4	System maps media files
FR-5	System renders chat UI
FR-6	User can play audio
FR-7	User can search messages
FR-8	System supports large chats
FR-9	No server upload required

9. Non-Functional Requirements

Requirement	Target
First render	< 5 seconds
Scroll performance	60 FPS
Audio latency	< 200ms
Memory optimization	required
Mobile responsive	yes

10. Future Features

- Multi-chat dashboard
- AI summarization
- Sentiment timeline
- Relationship analytics
- Export to PDF
- Timeline mode
- Media gallery
- Bookmarking

11. Monetization Ideas

- Premium export tools
- Cloud backup
- AI memory summaries
- PDF story generation
- Couple memory books